MSG No. 133.459 $P4_2/nbc$ [Type I, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,12,13,14]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[10,11,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,12,13,14]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,11,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,6,7,8]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,3,4,5]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9,14,15,16]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[10,11,12,13]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,6,10,11]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,9,14]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[4,5,15,16]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,8,12,13]`

Table 5: Wyckoff site: `8e`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,9]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,10]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,11]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[4,12]`
5	$\left[\frac{1}{2}, 0, 0\right]$	`[5,13]`
6	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,14]`
7	$\left[0, 0, \frac{1}{2}\right]$	`[7,15]`

8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[8,16]

Table 6: Wyckoff site: 8f, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[1,6]
2	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[2,3]
3	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	[4,5]
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[7,8]
5	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	[9,14]
6	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	[12,13]
8	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[1,6]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[2,3]
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[4,5]
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[7,8]
5	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[9,14]
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[12,13]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[15,16]

Table 8: Wyckoff site: 8h, site symmetry: .2.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	[1,4]
2	$\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$	[2,7]
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[3,8]
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	[5,6]
5	$\left[-x,\ \frac{3}{4},\ 0\right]$	[9,12]
6	$\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$	[10,15]
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[11,16]
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$	[13,14]

Table 9: Wyckoff site: 8i, site symmetry: .2.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[1,4]
2	$\left[\frac{1}{4},\ x,\ 0\right]$	[2,7]
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$	[3,8]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[5,6]
5	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[9,12]
6	$\left[\frac{3}{4},\ -x,\ 0\right]$	[10,15]
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$	[11,16]
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	[13,14]

Table 10: Wyckoff site: 8j, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	[1,7]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[2,5]
3	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[3,4]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[6,8]
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[9,15]
6	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[10,13]
7	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[11,12]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[14,16]

Table 11: Wyckoff site: 16k, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	[2]
3	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3]
4	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	[4]
5	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[5]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[6]
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[10]
11	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11]
12	$\left[-x,\ y+\frac{1}{2},\ z\right]$	[12]
13	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[13]
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[14]
15	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]